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Influence of Medical Versus Surgical Evacuation of Early Miscarriage Loss on Reproductive Outcomes of Women in Subsequent FET Cycle

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ABSTRACT

Purpose: To investigate how two methods for managing early miscarriage after frozen embryo transfer (FET) cycles impact live birth rates and other reproductive and perinatal outcomes in subsequent FET cycles without preimplantation genetic testing for aneuploidy (PGT-A).

Methods: This retrospective cohort study of women undergoing FET cycles (January 2016–December 2022) in our department who experienced early miscarriage diagnosed by transvaginal ultrasound examined the impact of medical versus surgical evacuation on subsequent live birth rates (LBR).

Results: Analysis of 1685 women revealed no significant differences in implantation, miscarriage, preterm birth, obstetric complications, or neonatal disease rates between groups. However, the surgical management group had lower positive pregnancy test, clinical pregnancy, and live birth rates, and higher cesarean rates. While adjustments for confounders eliminated the significance of differences in positive pregnancy tests and clinical pregnancies, lower live birth rates (aOR 0.80, 95% CI: 0.65–0.99) and higher cesarean rates (aOR 1.84, 95% CI: 1.19–2.84) persisted. The surgical group also showed significantly reduced endometrial thickness in subsequent cycles.

Conclusion: Subsequent FET cycles after surgical miscarriage evacuation show lower live birth rates and thinner endometrial lining than those following medical evacuation; surgical evacuation also correlates with increased cesarean section rates in non-PGT-A FET cycles.

1 | Introduction

Early Pregnancy Loss (EPL) is defined as a nonviable, intrauterine pregnancy diagnosed up to 12 weeks and 6 days gestation [1]. It is estimated that around 1 million pregnant individuals experience EPL in the United States annually, and globally, the incidence of miscarriage reaches approximately 23 million cases each year [2, 3]. Within the purview of IVF, emerging data from 2018 indicate that the miscarriage rate among infertile couples ranges between 19.3% and 21.4% [4].

There are three therapeutic options to tackle early pregnancy loss, namely expectant, medical, and surgical management. The latter two approaches are the most commonly adopted

Kunyao Deng and Jing Ye contributed equal to this work.

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in clinical practice [5]. The effectiveness, safety, and patient acceptance of medical and surgical approaches have been extensively discussed in previous studies [6–8]. For reproductive specialists, focus has been put on the impact of the management method of early pregnancy loss on subsequent fertility outcomes. It has been well established that the long- or short-term reproductive potential secondary to the medical versus surgical treatments for a miscarriage did not significantly differ in natural conception cycles [9, 10]. Nevertheless, reliable counseling on this subject for patients who previously did IVF and now are diagnosed with early miscarriage following a FET cycle remains challenging as evidence on which to guide this population is scarce and conflicting. Some investigators suggested that surgical evacuation could compromise future IVF outcomes based on the concept that the repeated dilatation and curettage procedure altered the uterine environment and decreased the endometrial thickness [11–13]. Conversely, other researchers failed to observe differences in pregnancy outcomes in subsequent embryo transfer cycles between women who received medical treatment or underwent surgical management for miscarriages in initial cycles [14]. Of note was that the above-listed literature was limited by a small sample size, which prevented them from drawing solid conclusions.

Despite encountering adverse pregnancy outcomes after prior FET cycles, a significant proportion of infertile women with early miscarriage persist in their profound aspiration for successful conception, seizing the maximal viable opportunity for attaining a live birth outcome. Information regarding how treatment options for miscarriage might affect reproductive performance in subsequent FET cycles is urgently needed for infertile patients to assist in their decision-making. Against this background, along with the ever-increasing adoption of IVF globally, the aim of the present study was therefore to survey the subsequent pregnancy outcomes after surgical versus medical treatment for early pregnancy loss following prior FET cycles.

2 | Materials and Methods

2.1 | Study Design and Patients

A retrospective study was performed at the Department of Assisted Reproduction of the Ninth People's Hospital of Shanghai Jiao Tong University School of Medicine. The present study compiled data on women who had suffered early miscarriages from January 2014 to December 2021 following FET cycles at our department and then included those who underwent the subsequent FET cycles from January 2016 to December 2022.

The exclusion criteria encompassed cases of biochemical pregnancies lacking evidence of gestational sacs, complete spontaneous abortions without any medical intervention, or suspected ectopic pregnancies. Patients opting out of subsequent FET cycles were also excluded. Additionally, women experiencing miscarriage following multifetal pregnancy reduction were excluded. Furthermore, the entire management of early miscarriage in this study was confined to a 5 day timeframe. If more than one miscarriage and subsequent embryo transfer cycles exist for the same woman in the database, only the first was retained. Approval for this study was granted by the Institutional Review Board of the hospital.

2.2 | Miscarriage Treatment Regimen

Our department employed specific criteria for the direct diagnosis of EPL utilizing transvaginal ultrasound scans during routine surveillance or in response to symptoms such as cramping and vaginal bleeding. The main criteria for a direct EPL diagnosis include (1) a mean gestational sac diameter of 25 mm or greater in the absence of an embryo; and (2) a crown-rump length of 7 mm or greater without detectable cardiac activity. In cases where patients present with signs suggestive of EPL but do not meet these definitive diagnostic criteria, our protocol mandates an additional ultrasound scan to be conducted after a 7 day interval for further assessment and diagnosis.

Women diagnosed with early miscarriage were treated with either medical evacuation or surgical evacuation. Surgical evacuation was typically performed within 1 week of EPL diagnosis or 24 h of medical treatment failure. For patients with a gestational age of less than 10 weeks, surgical evacuation was typically performed using vacuum aspiration with manual or electric suction devices, maintaining a negative pressure of 400–500 mmHg. For those with a gestational age between 10 and 12 weeks, forceps evacuation was applied to remove fetal and placental tissues, followed by vacuum aspiration to clear the remaining contents. Intravenous anesthesia was routinely administered, and 200 mg of oral doxycycline was given 1 h prior to the procedure for infection prophylaxis. If the evacuation was deemed incomplete based on visual inspection or ultrasound findings, an additional curettage was performed to ensure complete uterine clearance.

Since the year 2012, our hospital has been providing medical abortion services to EPL patients, regardless of their gestational age. The protocol was systematically established in our hospital, referring to research articles focused on medical abortions beyond 9 weeks of gestation in China [15]. The efficacy and safety of our protocol have been consistently demonstrated in clinical practice, leading us to maintain its utilization to date. To be specific, the protocol for medical evacuation varied depending on the diameter of the gestational sac. For patients with a gestational sac < 25 mm, the protocol included oral administration of 75 mg mifepristone for the initial 2 days (150 mg in total) and vaginal administration of 600mcg misoprostol at 6 a.m. on the third day. If no pregnancy tissue was passed, a second and possibly a third dose of 600 mcg misoprostol was administered vaginally at 10 a.m. and 2 p.m. For patients with a gestational sac > 25 mm, the dose of mifepristone was increased to 100 mg (200 mg in total), and the dose of misoprostol could sometimes be increased to 800mcg. If the pregnancy tissue had not passed, we would re-check the uterine cavity via transvaginal ultrasound at 8 p.m. on the third day to determine whether it had passed unconsciously.

2.3 | Data Source and Follow-Up of Pregnancy

Patient data for this study were extracted from the electronic records stored in our department's database. This database

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encompasses comprehensive and detailed data, including demographic and baseline characteristics of couples, medical and reproductive histories, indications for fertility treatment, cycle-specific treatment variables, laboratory parameters, pregnant outcomes, and perinatal outcomes [16]. Each couple was assigned a unique personal identification number, under which they systematically recorded and updated their treatment information from the initiation of the IVF procedure to the point of embryo transfer. The data collection and updating processes were continuously carried out by staff skilled in data management. Our follow-up protocols and any adverse outcome assessments have been elaborately discussed in our prior publications [17, 18].

2.4 | Endometrial Preparation and FET Protocols

Our facility utilized three types of protocols for endometrial preparation: modified natural cycles, stimulated cycles, and artificial cycles, as previously described [19–21]. To be specific, for patients with regular menstrual cycles, the initial approach to FET was a modified natural cycle. Starting from day 10 of the cycle, we employed serum hormone assays and transvaginal ultrasound to monitor follicular development, with follow-up assessments every 2 days. In cases of irregular menstrual cycles, a stimulated cycle was conducted by using letrozole, supplemented with human menopausal gonadotropin (hMG) when necessary. Ovulation was then induced using human chorionic gonadotropin (hCG) in the aforementioned two methods. In cases where patients presented with a history of thin endometrium in natural or stimulated cycles, and for those encountering logistical challenges, an artificial-cycle FET was employed. This method typically involved a sequential oral regimen of estradiol valerate followed by micronized vaginal progesterone. Continuation of progestin supplementation was maintained until the eighth week of gestation in cases where pregnancy was identified.

The vitrification and thawing methods employed in our study were previously described by Kuwayama et al. [22] Briefly, vitrification of cleavage-stage embryos and blastocysts was conducted using the Cryotop carrier system (Kitazato Biopharma Co.). The cryoprotectant solution consisted of 15% (v/v) ethylene glycol, 15% (v/v) dimethyl sulfoxide, and 0.5 M sucrose. Thawing involved the sequential use of cryoprotectant dilutions with sucrose concentrations of 1 M, 0.5 M, and 0 M. All procedures for vitrification and warming were performed at room temperature except for the first warming step, which was executed at 37°C.

It was important to highlight that PGT-A was not employed among the patients included in the study due to its unavailability in our department. Additionally, our center routinely implemented a nonelective freeze-all policy by vitrification. This protocol is adopted based on the evidence indicating that freeze-all methods offer pregnancy outcomes comparable to those of fresh embryo transfers [23], while also reducing the risk of ovarian hyperstimulation syndrome [24–26]. Moreover, in compliance with China's national guidelines, a maximum of two embryos was allowed to be transferred [27].

2.5 | Outcomes

In this study, the primary outcome measure focused on the live birth rate during FET cycles, examining various management approaches for early miscarriage. Live birth was specifically defined as the delivery of at least one viable baby after 28 weeks of gestation.

Secondary outcome measures included positive pregnancy test results, clinical pregnancies, implantation rates, miscarriage occurrences, alterations in endometrial thickness across consecutive FET cycles, and perinatal outcomes. A positive pregnancy test was identified as a serum hCG level exceeding 5 mIU/ml 14 days post-FET. Clinical pregnancy was affirmed through the detection of at least one gestational sac within the uterine cavity during ultrasound scanning at 5 weeks post-embryo transfer. The implantation rate was determined by the ratio of gestational sacs to the number of embryos transferred. Miscarriage was characterized as the spontaneous loss of a clinical pregnancy before reaching 20 weeks of gestation.

Endometrial thickness (EMT) was defined by the maximal distance from one interface of endometrial-myometrial to the other in the midsagittal plane. In natural cycle and mildly stimulated cycle FETs, EMT was taken from the day of hCG administration, while in artificial cycle FETs, EMT from the last ultrasound before progesterone initiation was used.

Perinatal outcomes evaluated were preterm birth, the mode of labor, pregnancy complications, and neonatal disease. Preterm birth was defined as birth before 37 weeks of gestation, and we used the rate of cesarean section to represent the mode of labor [28]. Pregnancy complications included those likely to happen in the perinatal period of pregnancy (i.e., gestational diabetes mellitus, hypertensive disorders of pregnancy including pregnancy-induced hypertension and pre-eclampsia, intrahepatic cholestasis of pregnancy, acute fatty liver of pregnancy, and abnormal placentation including placenta praevia and placental abruption) [29]. Neonatal disease in the study included congenital abnormality (i.e., polydactyly, syndactyly, and congenital heart disease) and other neonatal disorders (i.e., neonatal asphyxia, acute respiratory distress syndrome, pathologic jaundice, and meconium aspiration syndrome). Notably, only singleton live births were compared for perinatal outcomes.

2.6 | Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics (version 26.0; IBM Corp., Armonk, NY, USA) and R (version 4.3.1; R Foundation for Statistical Computing, Vienna, Austria). Specifically, R was used for inverse probability of treatment weighting (IPTW) and power analysis, while all other statistical procedures were conducted in SPSS. The chi-square test was utilized for the analysis of categorical, parametric variables. For the comparison of continuous variables that followed a normal distribution, two-sided t-tests were employed. Conversely, the Mann–Whitney U-test was applied for the comparison of continuous variables that were not normally distributed. A probability value of < 0.05 was considered

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statistically significant. Multivariate logistic regression analysis and propensity score matching were also performed. Analyses were adjusted for age, body mass index, infertility duration, prior history of miscarriage, prior history of surgical evacuation, parity, main infertility cause, FET cycle rank, fertilization method, endometrial preparation, interval between 2 cycles, gestational age of miscarriage cycle, endometrial thickness of miscarriage cycle, number of embryos transferred, embryo developmental stage, embryo quality at transfer of best embryo transferred, year of treatment. Power analysis showed that at least 746 patients would be needed (373 in each group) to have 80% power to detect a 10% difference in the primary outcomes.

A propensity score matching approach is considered to be superior to conventional regression-based methods when estimating treatment effects in nonrandomized observational studies. Therefore, to further verify the results, a PSM model was additionally applied to minimize selection bias and balance the baseline characteristics between groups. Propensity scores were estimated using logistic regression, incorporating the same set of potential confounders included in the aforementioned multivariable models. Specifically, "year of treatment" was used as an exact matching variable to account for time-dependent effects, while other covariates were entered as predictors in the logistic regression model to generate the propensity score. Subjects in the study group were matched at a 1:1 ratio to those in the reference controls via nearest neighbor matching, with a caliper width of 0.1. In addition, to further examine the robustness of the primary outcome, IPTW analysis was performed. Propensity scores were estimated using a nonparametric generalized boosted model (GBM), with the treatment group as the outcome variable and all aforementioned baseline covariates included in the model. Stabilized weights were calculated for the average treatment effect and were truncated at the 1st and 99th percentiles to limit the influence of extreme weights. To further assess potential effect modification, subgroup analyses were conducted stratified by four age categories (< 35, 35–37, 38–40, and > 40 years), and gestational age (< 10, and 10–12 weeks). A two-sided $p < 0.05$ was considered statistically significant.

3 | Results

3.1 | Patients' Demographics and Cycle Characteristics

In this study, we conducted a comprehensive review of 54,766 electronic files of FET cycles spanning from January 2016 to December 2022. There were 40 patients excluded from the study due to their inability to undergo embryo transfer in subsequent FET cycles, with 19 of them attributed to identified intrauterine diseases and 21 of them for other reasons, including personal preference, unforeseen fever before FET, or constraints imposed by working or commuting. The dataset comprised 1692 women who satisfied the inclusion criteria. Seven women, four from the medical evacuation group and three from the surgical evacuation group, were ultimately excluded as they had not reached the perinatal period by the study's conclusion. Consequently, the final dataset required for the study was available for 1685 women. Among these, 864 patients underwent medical evacuation, while 821 patients received surgical management. The inclusion and selection procedure is detailed in Figure 1. Patients' allocation to management groups was determined based on the final treatment strategy that they underwent.

The patient characteristics between the two groups (Table 1) did not exhibit any significant differences, except for the interval between the miscarriage cycle and the subsequent cycle, and the distribution of the treatment year. The interval between the 2 cycles for the medical group was significantly shorter than that for the surgical group (255.0(191.0, 387.8)d vs. 288.0(213.0, 420.5)d, respectively; $p < 0.001$). Another significant difference was observed in the distribution of the treatment year ($p < 0.001$). In the medical group, 46.4% of patients entered consecutive FET cycles after a miscarriage from 2016 to 2017; 23.0% of patients entered between 2018 and 2019, and 30.6% of patients entered from 2020 to 2022. In contrast, the distribution in the surgical group was 36.8%, 35.0%, and 28.3%, respectively.

3.2 | Reproductive Outcome Differences Between Surgical and Medical Evacuation Groups in Univariable and Multivariable Regression Analyses

In the univariable analysis (Table 2), no significant differences were observed in terms of implantation rates (OR, 0.87; 95% CI, 0.75–1.01), miscarriage rates (OR, 1.11; 95% CI, 0.75–1.66), preterm birth rates (OR, 0.78; 95% CI, 0.39–1.56), obstetric complications (OR, 0.79; 95% CI, 0.47–1.31), and neonatal diseases (OR, 0.60; 95% CI, 0.28–1.31). Excluding these factors, the remaining reproductive outcomes demonstrated statistically significant differences between the two groups. Specifically, patients who underwent surgical management exhibited lower rates of positive pregnancy tests, clinical pregnancy, and live births compared to those who opted for medical management. The rates of positive pregnancy tests, clinical pregnancy, and live birth were 55.2% versus 50.1% ($p = 0.034$, OR,0.81; 95% CI, 0.67–0.98), 50.5% versus 44.7% ($p = 0.018$, OR,0.79; 95% CI, 0.66–0.96), and 42.4% versus 35.9% ($p = 0.007$, OR,0.76; 95% CI, 0.63–0.93), respectively. Additionally, a significant difference was observed in the rate of cesarean sections ($p = 0.005$, OR,1.77; 95% CI, 1.78–2.65) between the two management groups.

After adjusting for key confounders (Table 2), the implantation (adjusted OR, 0.85; 95% CI, 0.69–1.03), miscarriage (adjusted OR, 1.03; 95% CI, 0.67–1.58), preterm birth (adjusted OR, 0.84; 95% CI, 0.39–1.81), obstetric complications (adjusted OR, 0.75; 95% CI, 0.42–1.31), and neonatal diseases (adjusted OR, 0.60; 95% CI, 0.24–1.46) remained comparable between the medical and surgical groups. However, the differences in positive pregnancy tests ($p = 0.111$, adjusted OR, 0.85; 95% CI, 0.69–1.04) and clinical pregnancies ($p = 0.081$, adjusted OR, 0.83; 95% CI, 0.68–1.02) between the two groups were no longer significant. Nonetheless, patients who underwent surgical evacuation during the miscarriage cycle continued to exhibit lower live birth rates ($p = 0.042$, adjusted OR, 0.80; 95% CI, 0.65–0.99) and higher cesarean section rates ($p = 0.006$, adjusted OR, 1.84; 95% CI, 1.19–2.84). In subgroup analyses stratified by age, the medical management

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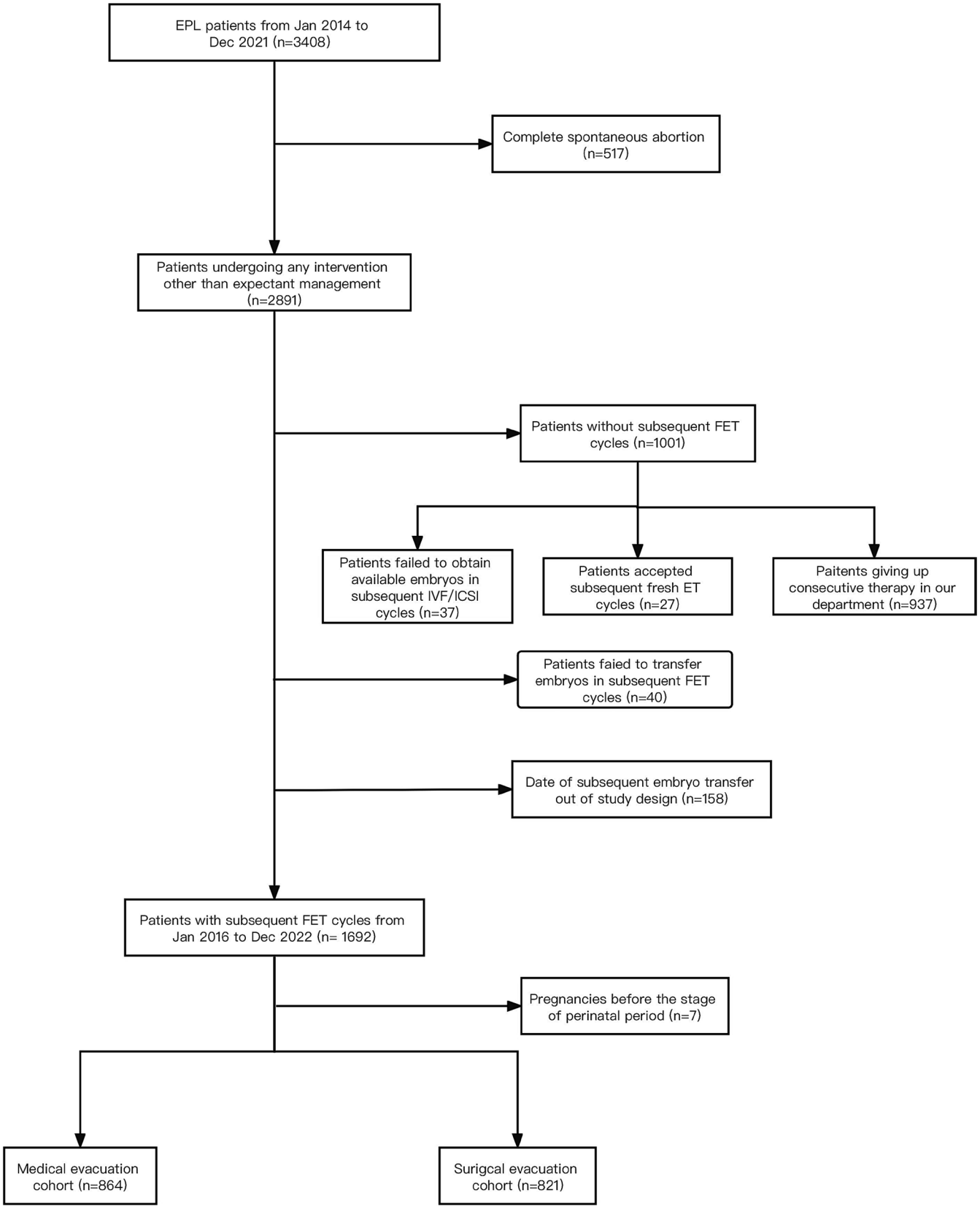


FIGURE 1 | Flowchart of the study design and phases.

group remained significantly associated with a higher live birth rate among women aged <35 years (aOR = 0.75; 95% CI: 0.56–0.99), whereas no significant differences were observed in other age groups (Table S1). The details of the gestational age-stratified analyses are presented in Tables S3, S4, indicating that variations in surgical techniques based on gestational age within the surgical group did not materially affect the main conclusions of the present study.

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TABLE 1 | Baseline and treatment characteristics of the study cohorts.

Group	Medical group	Surgical group	*p*
Patient	$n = 864$	$n = 821$	
Age at embryo transfer (years)	34.7 ± 5.1	34.8 ± 4.9	0.469[a]
BMI (kg/m^2)	22.2 ± 3.4	22.3 ± 3.2	0.690[a]
Infertility duration (years)	3.0 (1.0, 5.0)	3.0 (1.0, 5.0)	0.327[c]
Prior history of miscarriage			0.491[b]
0	738 (85.4)	692 (84.3)	
1	111 (12.8)	108 (13.2)	
≥ 2	15 (1.7)	21 (2.6)	
Prior history of surgical evacuation			0.523[b]
0	671 (77.7)	619 (75.4)	
1	127 (14.7)	130 (15.8)	
≥ 2	66 (7.6)	72 (8.8)	
Parity			0.120[b]
0	759 (87.8)	700 (85.3)	
≥ 1	105 (12.2)	121 (14.7)	
Main infertility cause*			0.289[b]
Tubal	277 (32.1)	260 (31.7)	
PCOS	97 (11.2)	75 (9.1)	
Endometriosis	74 (8.6)	66 (8.0)	
DOR	67 (7.8)	88 (10.7)	
Uterine	66 (7.6)	74 (9.0)	
Male	229 (26.5)	205 (25.0)	
Unexplained	54 (6.3)	53 (6.5)	
Gestational age of miscarriage cycle(d)	64.2 ± 8.5	64.4 ± 10.7	0.726[a]
Patients required another IVF cycle	227 (26.3)	226 (27.5)	0.562[b]
FET cycle rank			0.673[b]
2	483 (55.9)	452 (55.1)	
3	226(26.2)	208 (25.3)	
≥ 4	155 (17.9)	161 (19.6)	
Fertilization method			0.684[b]
IVF	485 (56.1)	459 (55.9)	
ICSI	296 (34.3)	273 (33.3)	
IVF + ICSI	83 (9.6)	89 (10.8)	
Endometrial preparation			0.649[b]
Modified natural cycle	289 (33.4)	262 (31.9)	
Stimulated cycles	234 (27.1)	238 (29.0)	
Hormonal replacement	341 (39.5)	321 (39.1)	
Interval between 2 cycles(d)	255.0 (191.0, 387.8)	288.0 (213.0, 420.5)	< 0.001[c]

(Continues)

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TABLE 1 | (Continued)

Group	Medical group	Surgical group	*p*
Endometrial thickness of miscarriage cycle(mm)	10.5 ± 2.2	10.5 ± 2.3	0.446[a]
Number of embryos transferred			0.083[b]
1	209 (24.2)	229 (27.9)	
2	655 (75.8)	592 (72.1)	
Embryo developmental stage			0.788[b]
Day 3	672 (77.8)	643 (78.3)	
Day 5/6	192 (22.2)	178 (21.7)	
Embryo quality at transfer of best embryo transferred (%)			0.627[b]
Grade 1 (good-quality)	800 (92.6)	755 (92.0)	
Grade 2 (poor-quality)	64 (7.4)	66 (8.0)	
Year of treatment*			< 0.001[b]
2016–2017	401 (46.4)	302 (36.8)	
2018–2019	199 (23.0)	287 (35.0)	
2020–2022	264 (30.6)	232 (28.3)	

Note: Normal distribution quantitative data are presented as mean ± SD. Non-normal distribution quantitative data are presented as median (first quartile, third quartile). Qualitative data are presented as *n* (%).
Abbreviations: BMI, body mass index; DOR, diminished ovarian reserve; FET, frozen embryo transfer; ICSI, intracytoplasmic sperm injection; IVF + ICSI, half IVF and half ICSI; IVF, in vitro fertilization; PCOS, polycystic ovary syndrome.
[a]Student's *t*-test.
[b]$\chi 2$ test.
[c]Mann–Whitney U-test.
*Percentages are rounded to one decimal place and may not total exactly 100%.

3.3 | The Effect of Miscarriage Management on Endometrial Thickness in the Consecutive Cycle

To figure out the possible factors exerting an influence on reproductive outcomes, we conducted a comprehensive analysis of endometrial thickness data. Our examination of endometrial thickness during miscarriage cycles did not reveal a statistically significant difference between the medical and surgical cohorts (10.5 ± 2.2 mm vs. 10.5 ± 2.3 mm, respectively; $p = 0.446$). However, a significant difference was observed in the endometrial thickness during subsequent cycles between the two groups (10.4 ± 2.2 mm vs. 9.8 ± 2.2 mm, respectively; $p < 0.001$). When we further scrutinized the average endometrial thickness variation for each group independently, the medical group did not exhibit a significant change (a decrease of 0.1 ± 2.1 mm in endometrial thickness, $p = 0.130$). In contrast, the surgical group demonstrated a statistically significant reduction (a decrease of 0.6 ± 2.3 mm in endometrial thickness; $p < 0.001$). The disparity in reduction between the two groups remained significant (0.1 ± 2.1 mm vs. 0.6 ± 2.3 mm, respectively; $p < 0.001$).

3.4 | Propensity Score Matching Analysis

Results from PSM are listed in Tables 3 and 4. All baseline demographics and cycle characteristics were no longer significantly different between the medical group and the surgical group after matching. Consistent with the results from the multivariate regression analysis, no differences were noted in terms of rates of positive pregnancy tests, clinical pregnancy, implantation, and miscarriage in the PSM models. Furthermore, the live birth rates remained significantly lower among women who accepted surgical treatment. Likewise, for the secondary outcomes, rates of preterm birth, pregnancy complications, and neonatal diseases remained statistically insignificant after PSM between the matched cohorts. Also, following the results from the multivariate regression analysis, rates of cesarean sections and outcomes in terms of endometrial thickness showed significant between-group differences.

3.5 | Inverse Probability of Treatment Weighting Analysis

The results of the IPTW analysis for live birth rates between the two groups are presented in Table 5, and the baseline characteristics of the "pseudo-randomized" sample are summarized in Table S2. All adjusted covariates showed absolute standardized differences of less than 0.1, indicating adequate balance across groups. After IPTW adjustment, live birth rates remained significantly lower among women who underwent surgical treatment ($p = 0.015$; adjusted OR, 0.76; 95% CI, 0.63–0.95).

4 | Discussion

The present study primarily explored the effects of two major miscarriage management approaches on live birth rates in the subsequent FET cycles and also analyzed their possible influence on other reproductive outcomes, endometrial thickness,

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TABLE 2 | Main reproductive outcomes.

Group	Medical group	Surgical group	OR (95% CI)	*p*	aOR (95% CI)	*p*
	n = 864	*n* = 821				
Positive pregnancy test	477 (55.2)	411 (50.1)	0.81 (0.67–0.98)	0.034	0.85 (0.69–1.04)	0.111
Implantation	546/1519	463/1413	0.87 (0.75–1.01)	0.070	0.85 (0.69–1.03)	0.095
Clinical pregnancy	436 (50.5)	367 (44.7)	0.79 (0.66–0.96)	0.018	0.83 (0.68–1.02)	0.081
Miscarriage	56/477 (11.7)	53/411 (12.9)	1.11 (0.75–1.66)	0.601	1.03 (0.67–1.58)	0.911
Live birth	366 (42.4)	295 (35.9)	0.76 (0.63–0.93)	0.007	0.80 (0.65–0.99)	0.042
Preterm birth	23/283 (8.1)	14/216 (6.5)	0.78 (0.39–1.56)	0.487	0.84 (0.39–1.81)	0.661
Cesarean section	188/283 (66.4)	168/216 (77.8)	1.77 (1.18–2.65)	0.005	1.84 (1.19–2.84)	0.006
Pregnancy complication	45/283 (15.9)	28/216 (13.0)	0.79 (0.47–1.31)	0.357	0.75(0.42–1.31)	0.308
Neonatal diseases	21/283 (7.4)	10/216 (4.6)	0.61 (0.28–1.31)	0.201	0.60 (0.24–1.46)	0.257
EM of subsequent cycle	10.4 ± 2.2	9.8 ± 2.2		< 0.001		
EM comparison of 2 cycles	0.1 ± 2.1	0.6 ± 2.3		< 0.001		

Note: Analyses were adjusted for age, BMI, infertility duration, prior history of miscarriage, prior history of surgical evacuation, parity, main infertility cause, gestational age of miscarriage cycle, FET cycle rank, fertilization method, endometrial preparation, interval between 2 cycles, endometrial thickness of miscarriage cycle, number of embryos transferred, embryo developmental stage, embryo quality at transfer of best embryo transferred, year of treatment.
Abbreviations: OR, odds ratios; aOR, adjusted odds ratios; CI, confidence interval; EM, endometrial thickness.

and perinatal outcomes. Our findings demonstrate that surgical evacuation following a FET cycle can compromise the live birth rate in the subsequent FET cycle.

Previous research among naturally conceiving women has extensively established the theory that reproductive outcomes are comparable between medical and surgical management, both in the short term and long term. A retrospective study conducted by Tam W.H. et al. detected a 13% and 12% difference in pregnancy rates in the first and second years, respectively [10]. However, they concluded that these differences were equivalent to cumulative pregnancy rates 5 years after either medical or surgical treatment for previous miscarriages. In another study, Graziosi et al. reported 126 early pregnancy losses and also showed a similar conception rate of 94% in the long term after either medical or surgical treatment for women [30]. Short-term effects were also examined in a recent observational cohort study, which reported comparable cumulative pregnancy rates within 1 year after different treatments [9]. However, it is important to note that this theory may not directly apply to infertile women due to population heterogeneity. Inherent differences in fecundity exist between IVF patients and naturally conceived women. The natural population possesses the ability to prepare for pregnancy monthly and multiple attempts at conception might mitigate the impact of a single dilation and curettage procedure on cumulative conception rates. On the contrary, although aiming to conceive as fast as possible and entirely dependent on ART, the infertile population was limited in the number of trials of embryo transfer and unable to reach 80% or even higher live birth rates in 1 year as previously reported among natural women. Therefore, our study shifted the focus from cumulative conception rates, commonly discussed in natural conception studies, to live birth rates in subsequent FET cycles following a miscarriage.

Limited existing literature has delved into the impact of diverse miscarriage management strategies on the reproductive outcomes in subsequent FET cycles, and the results were quite heterogeneous. Consistent with our study, Kemal Ozgur et al. revealed a significant adverse effect of surgical evacuation on live birth rates in the succeeding 6 months [31]. However, the study was constrained by a small sample size, and the reference group was composed of patients who had experienced biochemical pregnancy loss under 5 weeks. Another retrospective study conducted by Gilad Karavani et al. compared reproductive outcomes after three major management approaches, namely expectant management, medical management, and surgical evacuation, for first-trimester miscarriage [32]. Their findings indicated no significant difference in live birth rates in subsequent FET cycles across the three cohorts. Nevertheless, the medical group exhibited superior reproductive outcomes with elevated implantation and clinical pregnancy rates when compared to the surgical evacuation group. Despite these findings, this study reported alarmingly low pregnancy and live birth rates, at a mere 27% and 16% respectively. The study was also hampered by its limited sample size, with less than 100 patients in each cohort. In contrast to our findings, Meng et al. reported no significant difference in reproductive outcomes of subsequent embryo transfer cycles between patients who underwent surgical evacuation and those who did not [33]. However, this study did not clarify any detailed clinical strategies adopted in the reference group, namely, no surgical evacuation. The treatment protocols of neither medical nor expectant management were mentioned in this study. Moreover, this study did not use multivariable regression analyses or any other statistical approach to overcome potential bias in a retrospective study.

In terms of secondary outcomes, our study aligns with the findings from the aforementioned research with smaller sample sizes, indicating a significant decrease in endometrial

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TABLE 3 | Baseline and treatment characteristics of the study cohorts after propensity score matching.

Group	Medical group	Surgical group	*p*
Patient	*n* = 687	*n* = 687	
Age at embryo transfer (years)	34.7 ± 5.1	34.7 ± 5.0	0.979[a]
BMI (kg/m^2)	22.2 ± 3.3	22.2 ± 3.2	0.658[a]
Infertility duration (years)	3.0 (1.0, 5.0)	3.0 (1.0, 5.0)	0.469[c]
Prior history of miscarriage			0.828[b]
0	579 (84.3)	580 (84.4)	
1	94 (13.7)	90 (13.1)	
≥ 2	14 (2.0)	17 (2.5)	
Prior history of surgical evacuation*			0.853[b]
0	528 (76.9)	520 (75.7)	
1	103 (15.0)	106 (15.4)	
≥ 2	56 (8.2)	61 (8.9)	
Parity			0.380[b]
0	602 (87.6)	591 (86.0)	
≥ 1	85 (12.4)	96 (14.0)	
Main infertility cause*			0.429[b]
Tubal	222 (32.6)	211 (30.7)	
PCOS	79 (11.5)	65 (9.5)	
Endometriosis	56 (8.2)	54 (7.9)	
DOR	54 (7.9)	75 (10.9)	
Uterine	50 (7.3)	59 (8.6)	
Male	179 (26.1)	178 (25.9)	
Unexplained	45 (6.6)	42(6.6)	
Gestational age of miscarriage cycle(d)	64.3 ± 8.5	64.1 ± 10.5	0.774[a]
Patients required another IVF cycle	194 (28.2)	183 (26.6)	0.506[b]
FET cycle rank			0.948[b]
2	373 (54.3)	379 (55.2)	
3	184 (26.8)	180 (26.2)	
≥ 4	130 (18.9)	128 (18.6)	
Fertilization method			0.622[b]
IVF	387 (56.3)	383 (55.7)	
ICSI	232 (33.8)	225 (32.8)	
IVF + ICSI	68 (9.9)	79 (11.5)	
Endometrial preparation*			0.848[b]
Modified natural cycle	223 (32.5)	222 (32.3)	
Stimulated cycles	188 (27.4)	197 (28.7)	
Hormonal replacement	276 (40.2)	268 (39.0)	
Interval between 2 cycles (d)	266.0 (195.0, 409.0)	280.0 (211.0, 404.0)	0.073[c]

(Continues)

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TABLE 3 | (Continued)

Group	Medical group	Surgical group	*p*
Endometrial thickness of miscarriage cycle (mm)	10.5 ± 2.2	10.5 ± 2.3	0.943[a]
Number of embryos transferred			0.759[b]
1	178 (25.9)	183(26.6)	
2	509 (74.1)	504(73.4)	
Embryo developmental stage			0.948[b]
Day 3	536 (78.0)	537 (78.2)	
Day 5/6	151 (22.0)	150 (21.8)	
Embryo quality at transfer of best embryo transferred (%)			0.551[b]
Grade 1 (good-quality)	635 (92.4)	629 (91.6)	
Grade 2 (poor-quality)	52 (7.6)	58 (8.4)	
Year of treatment*			1.000[b]
2016–2017	296 (43.1)	296 (43.1)	
2018–2019	190 (27.7)	190 (27.7)	
2020–2022	201 (29.3)	201 (29.3)	

Note: Normal distribution quantitative data are presented as mean ± SD. Non-normal distribution quantitative data are presented as median (first quartile, third quartile). Qualitative data are presented as *n* (%).
Abbreviations: BMI, body mass index; DOR, diminished ovarian reserve; FET, frozen embryo transfer; ICSI, intracytoplasmic sperm injection; IVF + ICSI, half IVF and half ICSI; IVF, in vitro fertilization; PCOS, polycystic ovary syndrome.
[a]Student's t-test.
[b]$\chi 2$ test.
[c]Mann–Whitney U-test.
*Percentages are rounded to one decimal place and may not total exactly 100%.

TABLE 4 | Main reproductive outcomes after propensity score matching.

Group	Medical group	Surgical group	OR (95% CI)	*p*
	n = 687	*n* = 687		
Positive pregnancy test	378(55.0)	350(50.9)	0.85(0.69–1.05)	0.130
Implantation	431/1196	393/1191	0.87(0.74–1.04)	0.118
Clinical pregnancy	349(50.8)	313(45.6)	0.81(0.66–1.01)	0.052
Miscarriage	46/378(12.2)	41/350(11.7)	0.96(0.61–1.50)	0.850
Live birth	293(42.6)	256(37.3)	0.80(0.64–0.99)	0.042
Preterm birth	21/226(9.3)	13/188(6.9)	0.73(0.35–1.49)	0.380
Cesarean section	152/226(67.3)	147/188(78.2)	1.75(1.12–2.72)	0.013
Pregnancy complication	36/226(15.9)	25/188(13.3)	0.81(0.47–1.41)	0.452
Neonatal diseases	15/226(6.6)	7/188(3.7)	0.54(0.22–1.36)	0.188
EM of subsequent cycle	10.4 ± 2.2	9.8 ± 2.2		< 0.001
EM comparison of 2 cycles	0.1 ± 2.0	0.7 ± 2.3		< 0.001

Note: Matching without replacement was performed using propensity scores through the nearest neighbor random matching algorithm. Propensity scores were calculated using logistic regression based on Age, BMI, Infertility duration, Prior history of miscarriage, Prior history of surgical evacuation, Parity, Main infertility cause, Gestational age of miscarriage cycle, FET cycle rank, Fertilization method, Endometrial preparation, Interval between 2 cycles, Endometrial thickness of miscarriage cycle, Number of embryos transferred, Embryo developmental stage, Embryo quality at transfer of the best embryo transferred, Year of treatment.

thickness within the surgical group [32, 33]. It is essential to note that attributing a thinner endometrium as the primary causative factor for the lower live birth rates in the surgical group is not appropriate. Existing research on the relationship between endometrial thickness and reproductive outcomes has consistently shown notable declines in clinical pregnancy and live birth rates only when endometrial thickness is below 7 and 8 mm [34–36]. Therefore, although current studies suggest that surgical evacuation may impact the endometrial condition in subsequent FET cycles, a mere reduction in

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TABLE 5 | Comparison of the live birth rate between groups after inverse probability of treatment weighting.

Group	Live birth(%)	aOR (95% CI)	*p*
Surgical group	36.0(274.0/761.0)	0.76(0.63–0.95)	0.015
Medical group	42.0(340.4/809.8)		

Note: Inverse probability of treatment weighting was performed using propensity scores estimated with generalized boosted models (GBM). Propensity scores were modeled on the following baseline covariates: age, BMI, infertility duration, prior history of miscarriage, prior history of surgical evacuation, parity, main infertility cause, gestational age of miscarriage cycle, FET cycle rank, fertilization method, endometrial preparation, interval between 2 cycles, endometrial thickness of miscarriage cycle, number of embryos transferred, embryo developmental stage, embryo quality at transfer of best embryo transferred, year of treatment. Stabilized weights for the average treatment effect (ATE) were computed and truncated at the 1st and 99th percentiles to limit extreme weights.

endometrial thickness alone does not sufficiently impact live birth outcomes, with the specific mechanisms remaining complex and unclear. Acknowledging the limited clinical impact of reduced endometrial thickness following a single surgical evacuation procedure, we still deem it an important piece of information to present. Considering previous literature indicating a possibly cumulative effect of surgical evacuation [11–13] on uterine endometrium, this finding is particularly significant for patients whose endometrial thickness is near or below the critical 7–8 mm threshold and for those dealing with recurrent miscarriages and frequently facing decisions about miscarriage management.

To the best of our knowledge, this study is the first to explore perinatal outcomes in subsequent FET cycles following varied strategies for miscarriage management. We observed a statistically significant increase in cesarean section rates post-surgical evacuation. Sandall et al. reported that nearly every woman undergoing a cesarean section faced an elevated risk of certain morbidities in subsequent pregnancies, and cesarean sections may also lead to both immediate and long-term detrimental effects on offsprings [28]. The notably higher cesarean section rates in the surgical group, while concerning, cannot be immediately attributed to surgical evacuation as the causative factor for increased cesarean deliveries. Any attempt to draw a direct causal link between the mode of miscarriage management and subsequent delivery mode should be made with great caution. It is possible that patients who chose surgical evacuation were inherently more inclined to select cesarean section as their preferred mode of delivery, irrespective of medical necessity. Additionally, to determine whether the difference arises from underlying medical causes, more comprehensive obstetric data, particularly regarding the clinical indications for cesarean delivery, would be required. Therefore, the present study is unable to conclusively establish whether surgical miscarriage management is independently associated with an increased likelihood of cesarean delivery in future pregnancies. Nevertheless, this observation warrants further in-depth investigation in future research.

The main limitation of the present study was a retrospective design, with inherent bias, and patients were categorized in terms of the final treatment they underwent instead of their initial options. Furthermore, an additional limitation was the lack of some important data, such as the size of the gestational sac and the crown-rump length. Moreover, although the World Health Organization (WHO) and the American College of Obstetricians and Gynecologists (ACOG) recommend suction methods over sharp curettage for surgical evacuation, the retrospective nature of our study precluded identification of the specific technique used in each case [1, 37]. Future prospective studies are warranted to determine whether different surgical methods affect reproductive outcomes. In addition, the majority of patients received two cleavage-stage embryos instead of a single blastocyst transfer, and our department adopted a freeze-all strategy, which may limit the generalizability of this study. Besides, the distribution of the year of treatment was significantly uneven between the two cohorts. Nevertheless, multivariable regression analyses did not show that the year of treatment was significantly correlated with live birth rates or any other outcomes in the subsequent FET cycle. Finally, the evacuation procedure in this study was not limited to the same hospital, which may cause variation.

The major strength of the present study lies in its relatively substantial and comparable sample size. Additionally, the uniformity in the protocol for the majority of patients in the medical group within our hospital bolstered the reliability of our findings. Instead of merely evaluating the live birth rates, this study also encompassed the assessment of other reproductive outcomes, endometrial thickness variations, and perinatal outcomes, thus making the results comprehensive. Moreover, the application of PSM and IPTW models further enhanced the robustness of our results by balancing clinically relevant demographic differences. The consistency of findings between the main analyses and the PSM models added additional credibility to our conclusions.

5 | Conclusions

In summary, the present study revealed that the live birth rates in the subsequent FET cycles after surgical evacuation of early miscarriages were lower compared to those after medical evacuation. This discovery could be instrumental in guiding miscarriage management strategies after FET cycles. Although the present study's evidence remains limited and might not warrant immediate changes in clinical practice, it offers important insights for future prospective research.

Ethics Statement

The study received approval from the hospital's Institutional Review Board, and due to its retrospective nature and anonymized data analysis, written consent was not required (no. SH9H-2023-T469-1).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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References

1. ACOG Practice Bulletin No. 200: Early Pregnancy Loss," *Obstetrics and Gynecology* 132, no. 5 (2018): 197–207, https://doi.org/10.1097/aog.0000000000002899.

2. S. Neill, "Management of Early Pregnancy Loss," *JAMA* 329, no. 16 (2023): 1399–1400, https://doi.org/10.1001/jama.2023.0933.

3. S. Quenby, I. D. Gallos, R. K. Dhillon-Smith, et al., "Miscarriage Matters: The Epidemiological, Physical, Psychological, and Economic Costs of Early Pregnancy Loss," *Lancet* 397, no. 10285 (2021): 1658–1667, https://doi.org/10.1016/s0140-6736(21)00682-6.

4. C. Wyns, C. De Geyter, C. Calhaz-Jorge, et al., "ART in Europe, 2018: Results Generated From European Registries by ESHRE," *Human Reproduction Open* 2022, no. 3 (2022): hoac022, https://doi.org/10.1093/hropen/hoac022.

5. Z. F. Meisel and C. A. Schreiber, "Variations in Care for Early Pregnancy Loss Across Clinical Settings," *JAMA Network Open* 6, no. 3 (2023): e232645, https://doi.org/10.1001/jamanetworkopen.2023.2645.

6. P. W. Ashok, A. Kidd, G. M. Flett, A. Fitzmaurice, W. Graham, and A. Templeton, "A Randomized Comparison of Medical Abortion and Surgical Vacuum Aspiration at 10-13 Weeks Gestation," *Human Reproduction* 17, no. 1 (2002): 92–98, https://doi.org/10.1093/humrep/17.1.92.

7. J. Zhang, J. M. Gilles, K. Barnhart, M. D. Creinin, C. Westhoff, and M. M. Frederick, "A Comparison of Medical Management With Misoprostol and Surgical Management for Early Pregnancy Failure," *New England Journal of Medicine* 353, no. 8 (2005): 761–769, https://doi.org/10.1056/NEJMoa044064.

8. T. Kelly, J. Suddes, D. Howel, J. Hewison, and S. Robson, "Comparing Medical Versus Surgical Termination of Pregnancy at 13-20 Weeks of Gestation: A Randomised Controlled Trial," *BJOG* 117, no. 12 (2010): 1512–1520, https://doi.org/10.1111/j.1471-0528.2010.02712.x.

9. Y. Tzur, O. Samueloff, Y. Raz, S. Bar-On, I. Laskov, and T. Tzur, "Conception Rates After Medical Versus Surgical Evacuation of Early Miscarriage," *Fertility and Sterility* 115, no. 1 (2021): 118–124, https://doi.org/10.1016/j.fertnstert.2020.07.016.

10. W. H. Tam, M. H. Tsui, I. H. Lok, S. K. Yip, P. M. Yuen, and T. K. Chung, "Long-Term Reproductive Outcome Subsequent to Medical Versus Surgical Treatment for Miscarriage," *Human Reproduction* 20, no. 12 (2005): 3355–3359, https://doi.org/10.1093/humrep/dei257.

11. A. B. Hooker, M. Lemmers, A. L. Thurkow, et al., "Systematic Review and Meta-Analysis of Intrauterine Adhesions After Miscarriage: Prevalence, Risk Factors and Long-Term Reproductive Outcome," *Human Reproduction Update* 20, no. 2 (2014): 262–278, https://doi.org/10.1093/humupd/dmt045.

12. A. Azumaguchi, H. Henmi, H. Ohnishi, T. Endo, and T. Saito, "Role of Dilatation and Curettage Performed for Spontaneous or Induced Abortion in the Etiology of Endometrial Thinning," *Journal of Obstetrics and Gynaecology Research* 43, no. 3 (2017): 523–529, https://doi.org/10.1111/jog.13254.

13. Y. Wang, Y. Sun, W. Di, Y. P. Kuang, and B. Xu, "Association Between Induced Abortion History and Later In Vitro Fertilization Outcomes," *International Journal of Gynaecology and Obstetrics* 141, no. 3 (2018): 321–326, https://doi.org/10.1002/ijgo.12481.

14. R. Tamir, S. Allouche, A. Weissman, et al., "The Effect of Medical Versus Surgical Treatment of Spontaneous Miscarriage on Subsequent In Vitro Fertilization Cycles," *Gynecological Endocrinology* 32, no. 3 (2016): 231–233, https://doi.org/10.3109/09513590.2015.1110693.

15. Q. J. Chen, S. P. Hou, C. Meads, et al., "Mifepristone in Combination With Prostaglandins for Termination of 10-16 Weeks' Gestation: A Systematic Review," *European Journal of Obstetrics, Gynecology, and Reproductive Biology* 159, no. 2 (2011): 247–254, https://doi.org/10.1016/j.ejogrb.2011.06.034.

16. X. Yang, J. Zhang, J. Wu, et al., "Association Between the Number of Oocytes Retrieved and Neonatal Outcomes After Freeze-All IVF Cycles," *Human Reproduction* 34, no. 10 (2019): 1937–1947, https://doi.org/10.1093/humrep/dez137.

17. T. Du, Y. Wang, Y. Fan, S. Zhang, Z. Yan, and W. Yu, "Fertility and Neonatal Outcomes of Embryos Achieving Blastulation on Day 7: Are They of Clinical Value?," *Human Reproduction* 33, no. 6 (2018): 1038–1051, https://doi.org/10.1093/humrep/dey092.

18. H. Chen, Y. Wang, Q. Lyu, et al., "Comparison of Live-Birth Defects After Luteal-Phase Ovarian Stimulation vs. Conventional Ovarian Stimulation for In Vitro Fertilization and Vitrified Embryo Transfer Cycles," *Fertility and Sterility* 103, no. 5 (2015): 1194–201.e2, https://doi.org/10.1016/j.fertnstert.2015.02.020.

19. T. Du, H. Chen, R. Fu, Q. Chen, Y. Wang, and B. W. Mol, "Comparison of Ectopic Pregnancy Risk Among Transfers of Embryos Vitrified on Day 3, Day 5, and Day 6," *Fertility and Sterility* 108, no. 1 (2017): 108–16.e1, https://doi.org/10.1016/j.fertnstert.2017.05.027.

20. J. Zhang, H. Liu, X. Mao, et al., "Effect of Endometrial Thickness on Birthweight in Frozen Embryo Transfer Cycles: An Analysis Including 6181 Singleton Newborns," *Human Reproduction* 34, no. 9 (2019): 1707–1715, https://doi.org/10.1093/humrep/dez103.

21. D. Chen, Q. Xu, X. Mao, J. Zhang, and L. Wu, "Obstetric and Perinatal Outcomes After Embryos Cultured in One-Step Versus Sequential Culture Media Systems in Vitrified-Warmed Single Blastocyst Transfer Cycles," *Reproductive Biomedicine Online* 47, no. 3 (2023): 103227, https://doi.org/10.1016/j.rbmo.2023.04.018.

22. M. Kuwayama, G. Vajta, O. Kato, and S. P. Leibo, "Highly Efficient Vitrification Method for Cryopreservation of Human Oocytes," *Reproductive Biomedicine Online* 11, no. 3 (2005): 300–308, https://doi.org/10.1016/s1472-6483(10)60837-1.

23. Q. Zhu, Q. Chen, L. Wang, et al., "Live Birth Rates in the First Complete IVF Cycle Among 20 687 Women Using a Freeze-All Strategy," *Human Reproduction* 33, no. 5 (2018): 924–929, https://doi.org/10.1093/humrep/dey044.

24. Z. J. Chen, Y. Shi, Y. Sun, et al., "Fresh Versus Frozen Embryos for Infertility in the Polycystic Ovary Syndrome," *New England Journal of Medicine* 375, no. 6 (2016): 523–533, https://doi.org/10.1056/NEJMoa1513873.

25. L. N. Vuong, V. Q. Dang, T. M. Ho, et al., "IVF Transfer of Fresh or Frozen Embryos in Women Without Polycystic Ovaries," *New England Journal of Medicine* 378, no. 2 (2018): 137–147, https://doi.org/10.1056/NEJMoa1703768.

26. T. Zaat, M. Zagers, F. Mol, M. Goddijn, M. van Wely, and S. Mastenbroek, "Fresh Versus Frozen Embryo Transfers in Assisted Reproduction," *Cochrane Database of Systematic Reviews* 2, no. 2 (2021): Cd011184, https://doi.org/10.1002/14651858.CD011184.pub3.

27. B. Huang, D. Hu, K. Qian, et al., "Is Frozen Embryo Transfer Cycle Associated With a Significantly Lower Incidence of Ectopic Pregnancy? An Analysis of More Than 30,000 Cycles," *Fertility and Sterility* 102, no. 5 (2014): 1345–1349, https://doi.org/10.1016/j.fertnstert.2014.07.1245.

28. J. Sandall, R. M. Tribe, L. Avery, et al., "Short-Term and Long-Term Effects of Caesarean Section on the Health of Women and Children," *Lancet* 392, no. 10155 (2018): 1349–1357, https://doi.org/10.1016/s0140-6736(18)31930-5.

29. Y. Shi, Y. Sun, C. Hao, et al., "Transfer of Fresh Versus Frozen Embryos in Ovulatory Women," *New England Journal of Medicine* 378, no. 2 (2018): 126–136, https://doi.org/10.1056/NEJMoa1705334.

30. G. C. Graziosi, H. W. Bruinse, P. J. Reuwer, O. Teteringen, and B. W. Mol, "Fertility Outcome After a Randomized Trial Comparing Curettage With Misoprostol for Treatment of Early Pregnancy Failure,"

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Human Reproduction 20, no. 6 (2005): 1749–1750, https://doi.org/10.1093/humrep/deh754.

31. K. Ozgur, H. Bulut, M. Berkkanoglu, F. O. Basegmez, and K. Coetzee, "Six-Month Recovery Needed After Dilation and Curettage (D and C) for Reproductive Outcomes in Frozen Embryo Transfer," *Journal of Obstetrics and Gynaecology* 38, no. 8 (2018): 1150–1157, https://doi.org/10.1080/01443615.2018.1460335.

32. G. Karavani, H. Alexandroni, D. Sheinin, et al., "Endometrial Thickness Following Early Miscarriage in IVF Patients - Is There a Preferred Management Approach?," *Reproductive Biology and Endocrinology* 19, no. 1 (2021): 93, https://doi.org/10.1186/s12958-021-00780-7.

33. J. Meng, M. Zhu, W. Shen, X. Huang, H. Sun, and J. Zhou, "Influence of Surgical Evacuation on Pregnancy Outcomes of Subsequent Embryo Transfer Cycle Following Miscarriage in an Initial IVF Cycle: A Retrospective Cohort Study," *BMC Pregnancy and Childbirth* 19, no. 1 (2019): 409, https://doi.org/10.1186/s12884-019-2543-9.

34. A. Kasius, J. G. Smit, H. L. Torrance, et al., "Endometrial Thickness and Pregnancy Rates After IVF: A Systematic Review and Meta-Analysis," *Human Reproduction Update* 20, no. 4 (2014): 530–541, https://doi.org/10.1093/humupd/dmu011.

35. K. E. Liu, M. Hartman, A. Hartman, Z. C. Luo, and N. Mahutte, "The Impact of a Thin Endometrial Lining on Fresh and Frozen-Thaw IVF Outcomes: An Analysis of Over 40 000 Embryo Transfers," *Human Reproduction* 33, no. 10 (2018): 1883–1888, https://doi.org/10.1093/humrep/dey281.

36. S. Hellerstein, S. Feldman, and T. Duan, "China's 50% Caesarean Delivery Rate: Is It Too High?," *BJOG* 122, no. 2 (2015): 160–164, https://doi.org/10.1111/1471-0528.12971.

37. World Health Organization, *Clinical Practice Handbook for Safe Abortion* (WHO, 2014), https://apps.who.int/iris/handle/10665/97415.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Crude and adjusted odds ratios for live birth rate across age subgroups. **Table S2:** Summary of baseline covariates in the IPTW-adjusted cohorts. **Table S3:** Crude and adjusted odds ratios for live birth rate across gestational-age subgroups in the surgical group. **Table S4:** Crude and adjusted odds ratios for live birth rate between medical and surgical groups across gestational age subgroups.

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